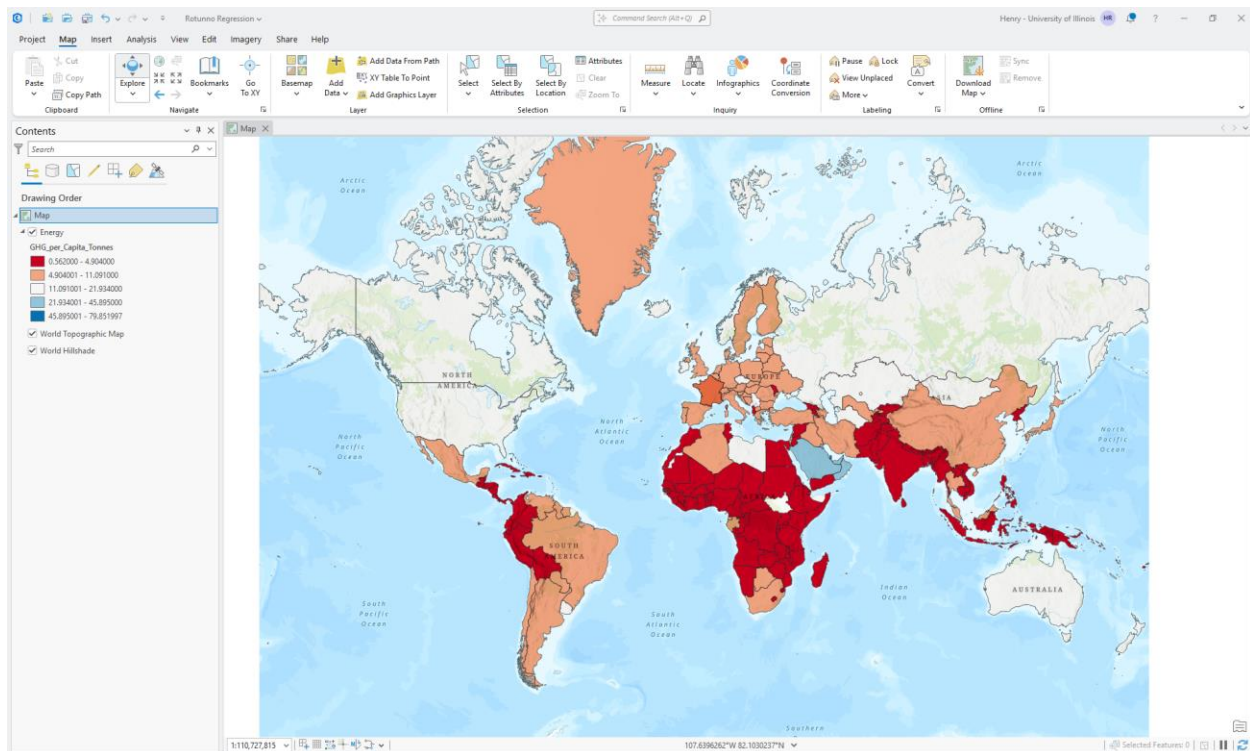


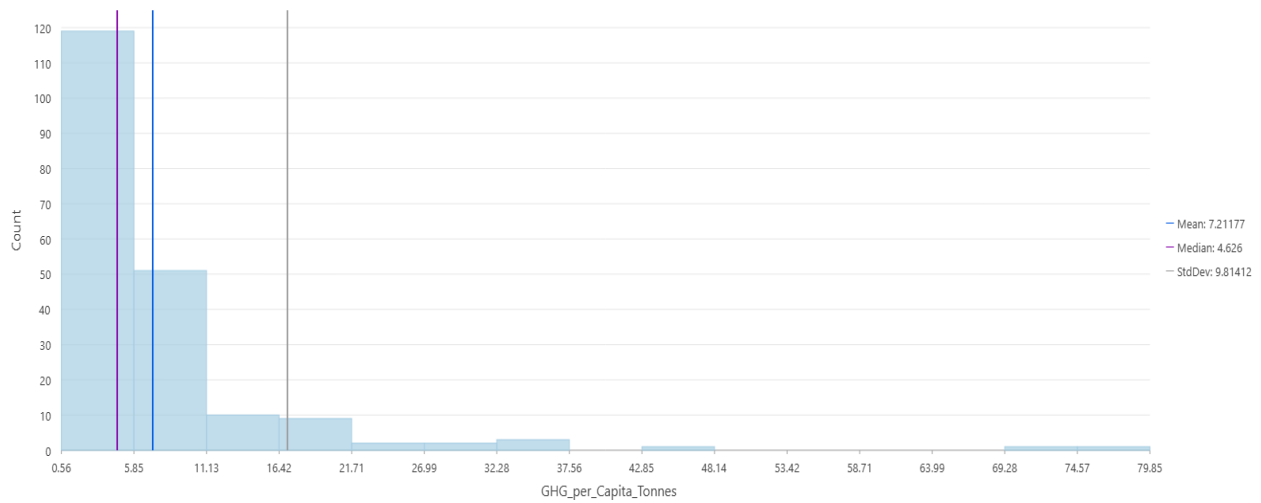
Regression

Henry Rotunno – Updated 26 February 2026

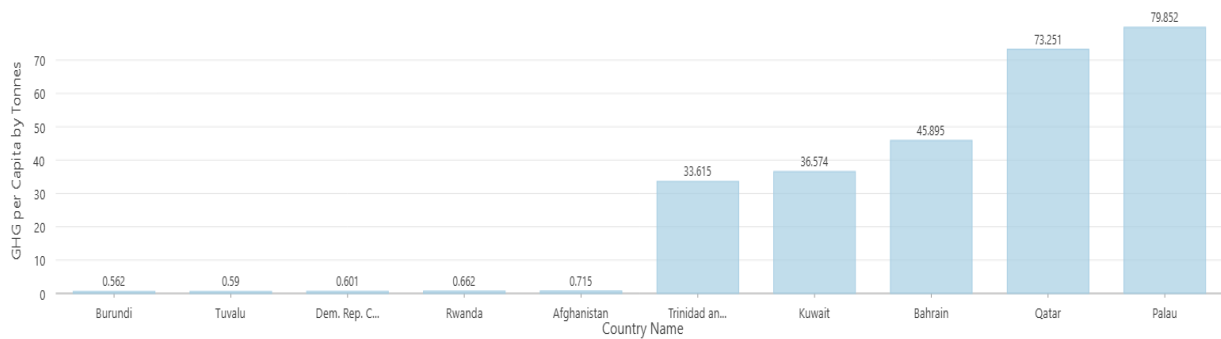
This report uses multiple regression to analyze the relationship between greenhouse gas emission per capita, coal consumption per capita, GDP per capita, gas consumption per capita, and percent renewable.



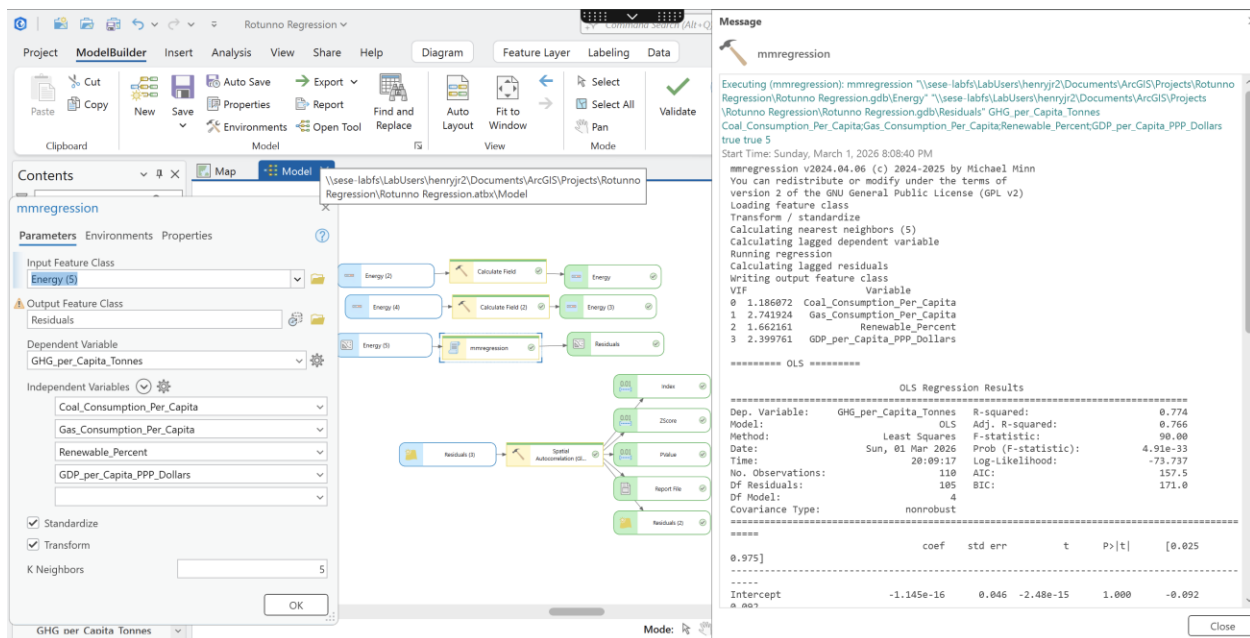
Greenhouse gas emissions in tonnes for each country in 2023 per capita (The World Bank n.d.)



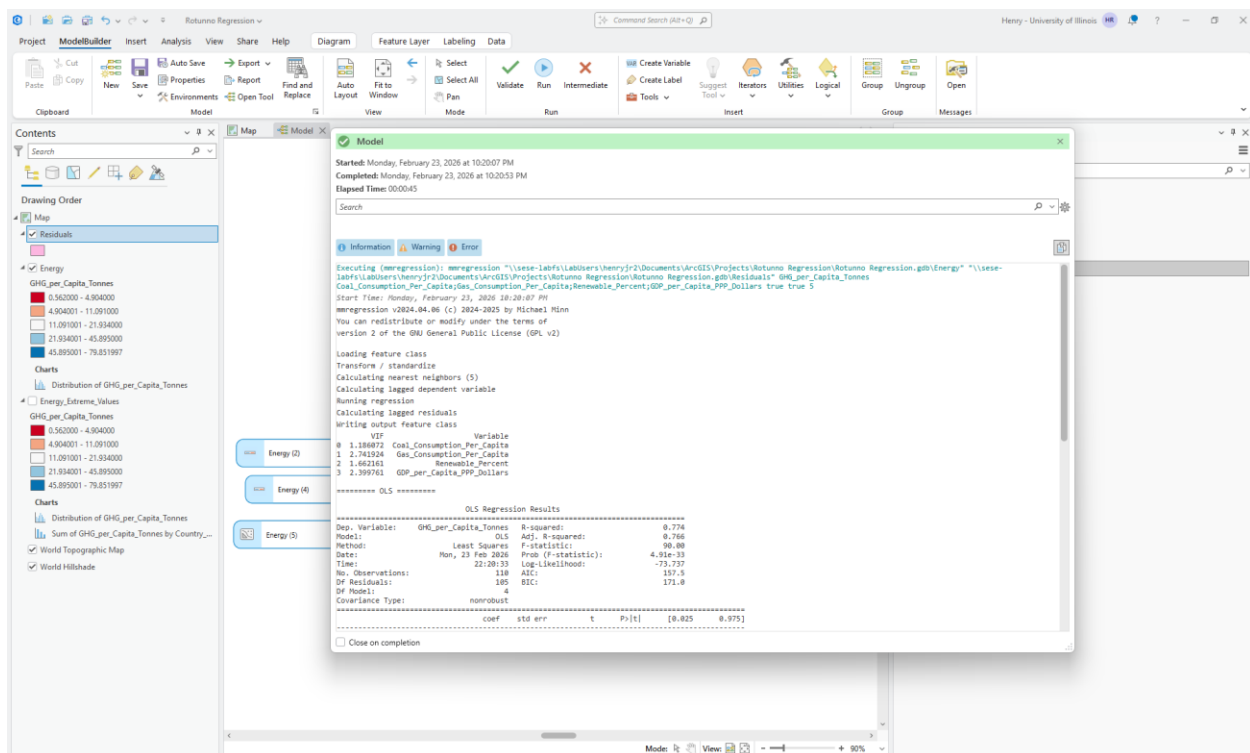
Right skewed distribution with a median of 4.626, minimum of 0.562, and maximum of 79.852 for greenhouse gasses per capita by country in tonnes in 2023.



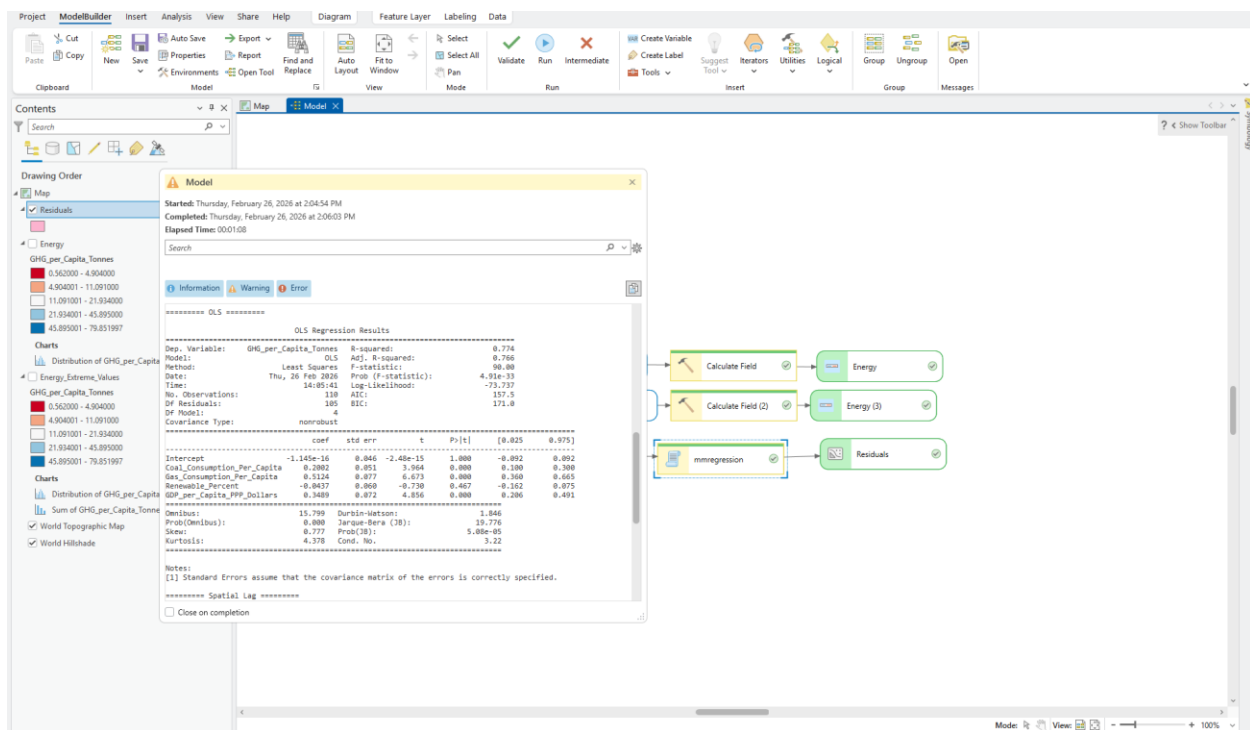
Palau has the highest greenhouse gas emissions per capita at 79.852 tonnes while Burundi has the lowest at 0.562 tonnes.



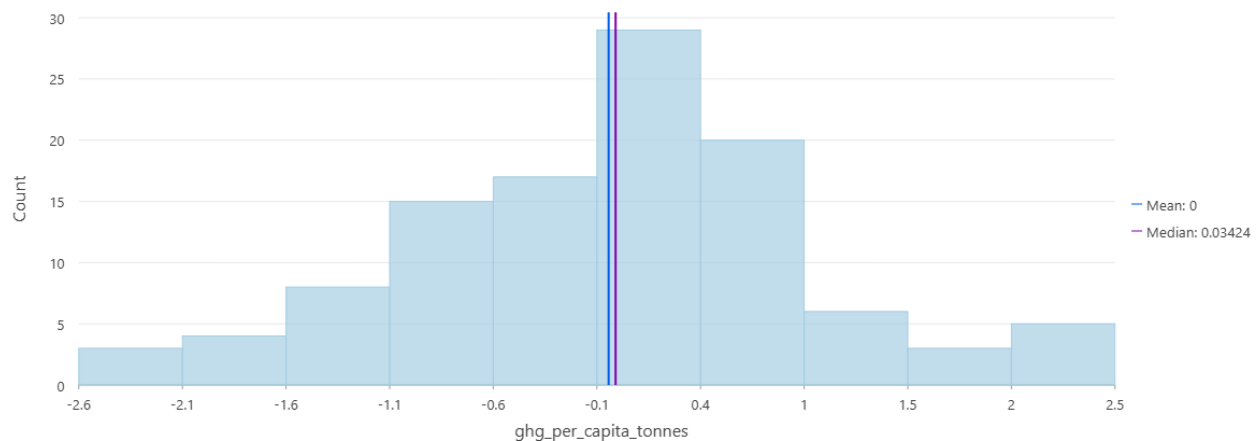
mmregression



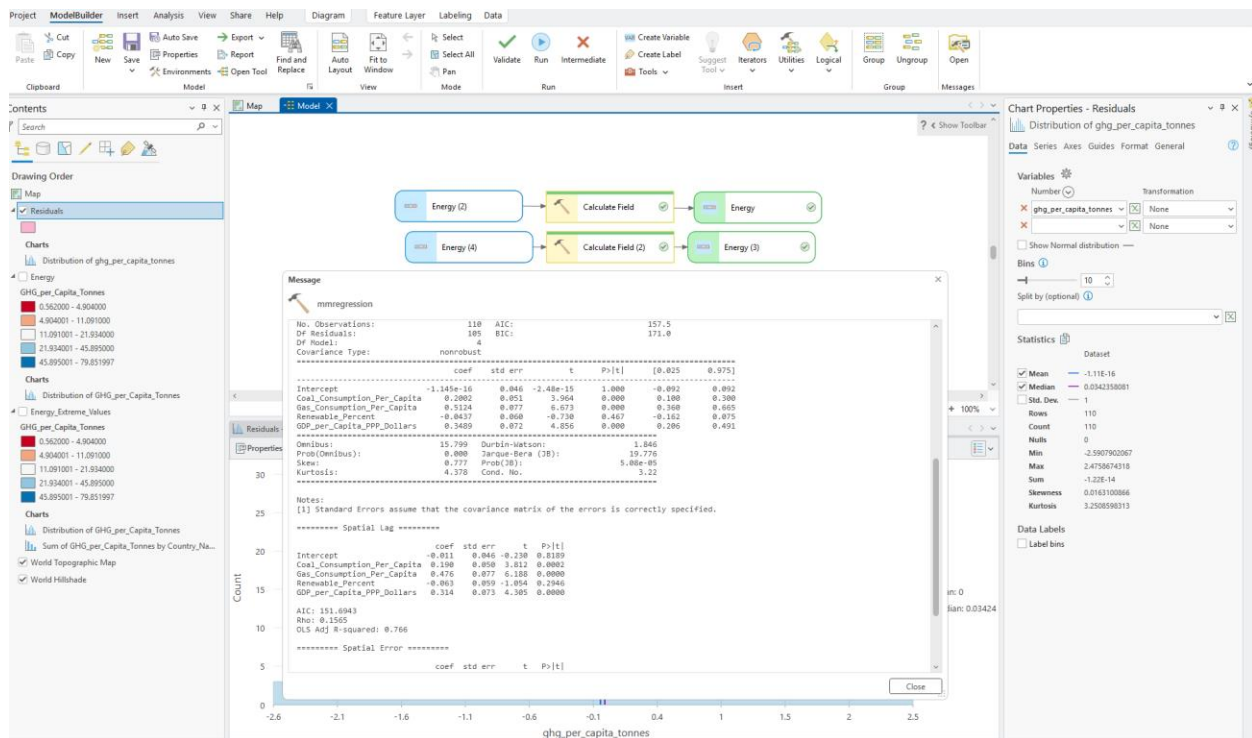
Gas consumption per capita has the highest multicollinearity with a VIF value of 2.74, consequently the model is not affected by multicollinearity.



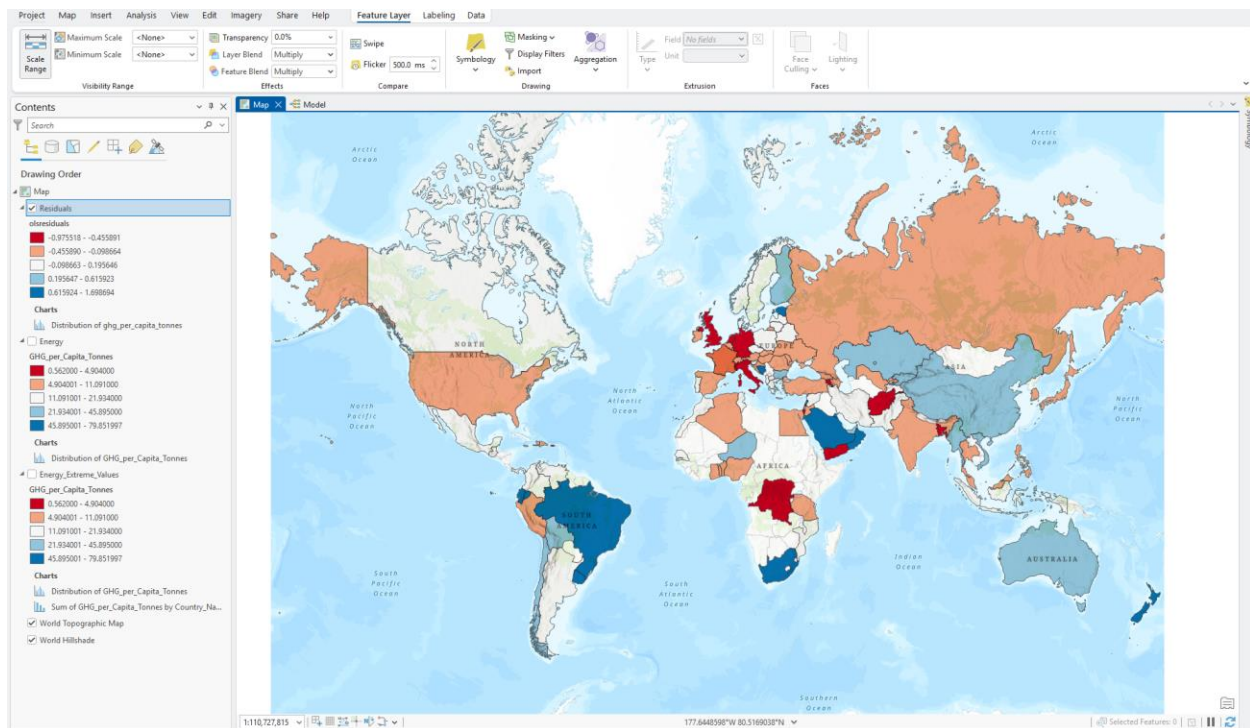
My R-squared value of 0.766 indicates that the model predicts 76.6% of the variance in greenhouse gas emissions in tonnes per capita.



Normal distribution for greenhouse gasses per capita in tonnes.

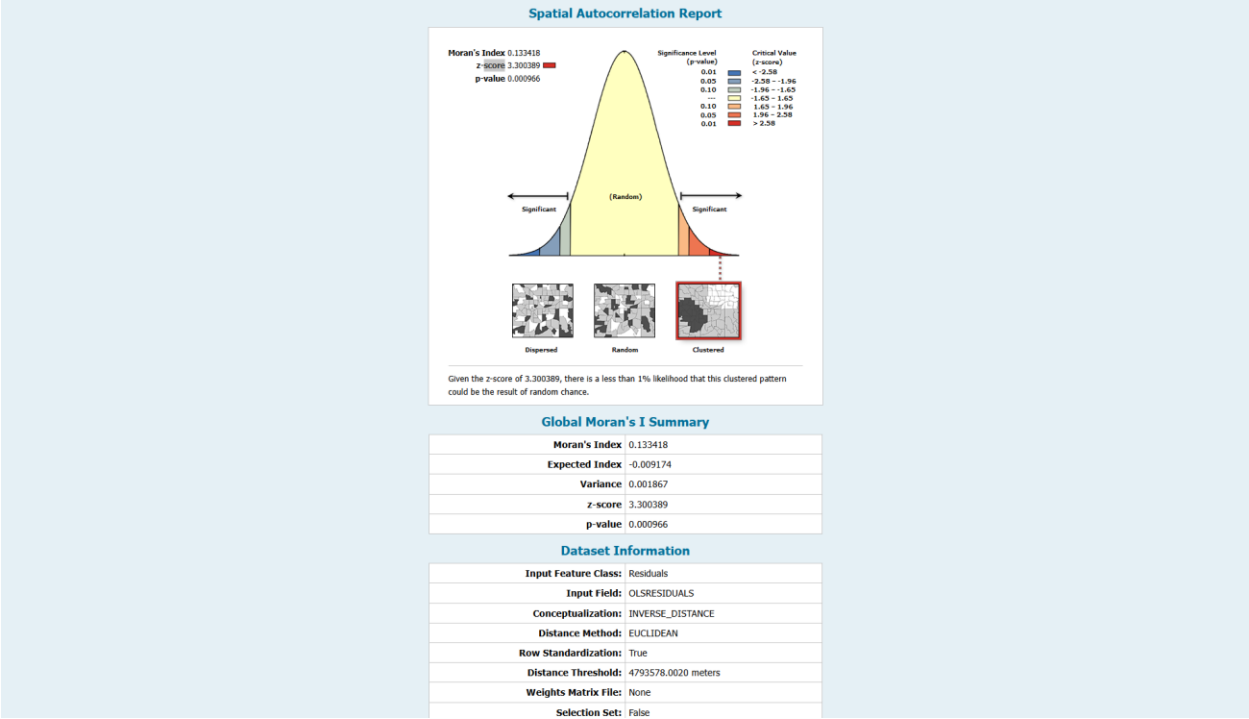


The OLS regression results indicate that gas consumption per capita's coefficient value of 0.5124 is the most important while percent renewable coefficient value of -0.0437 is the least significant.

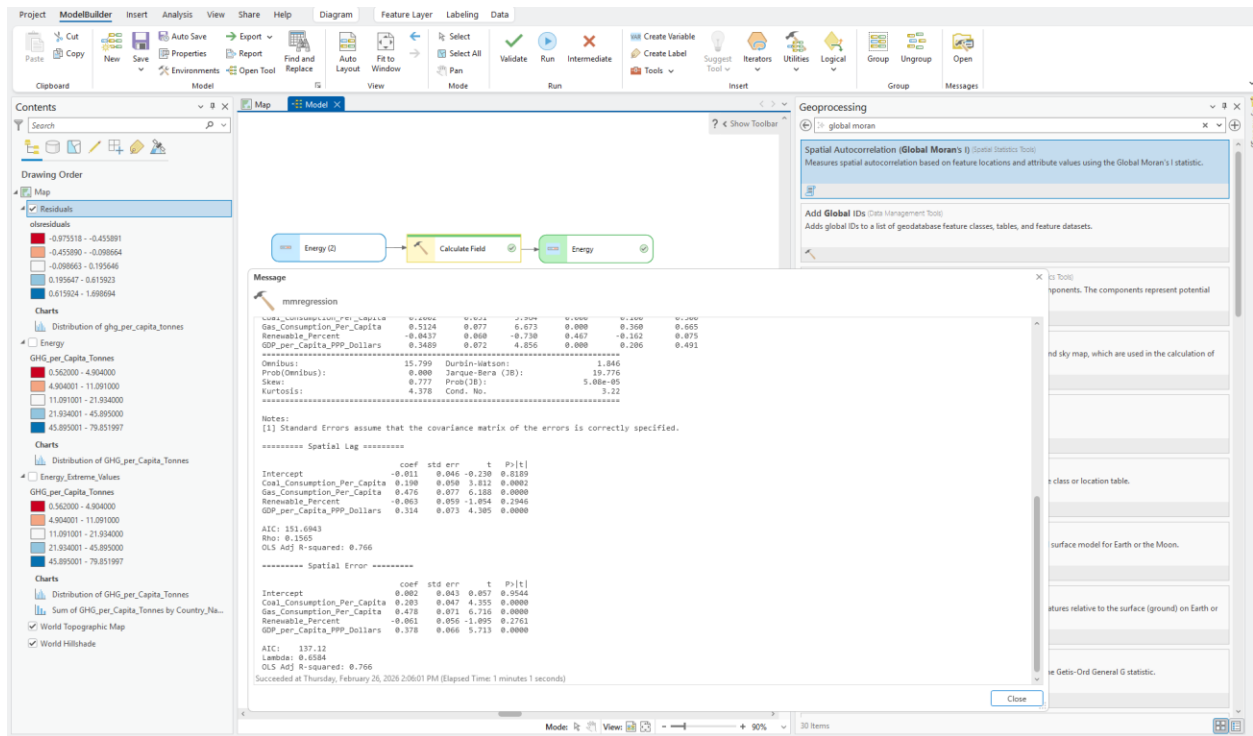


OLS Residuals

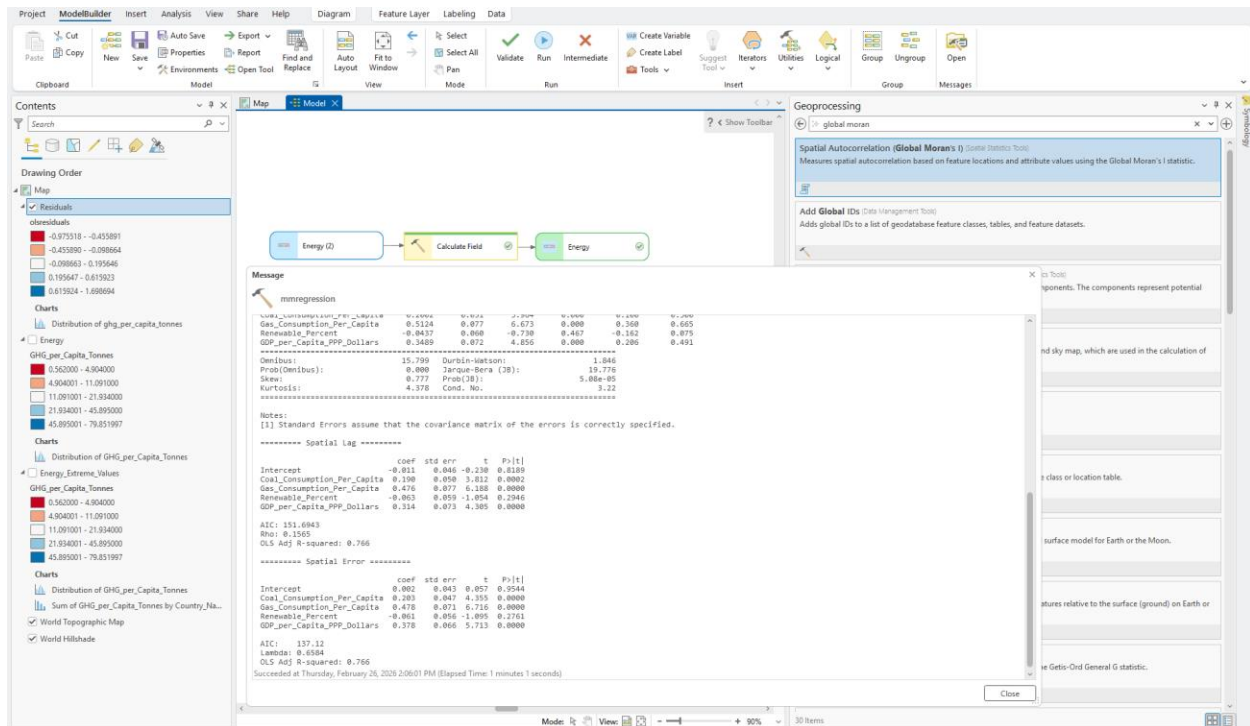
The model predicted values higher than the actual values in western Europe.



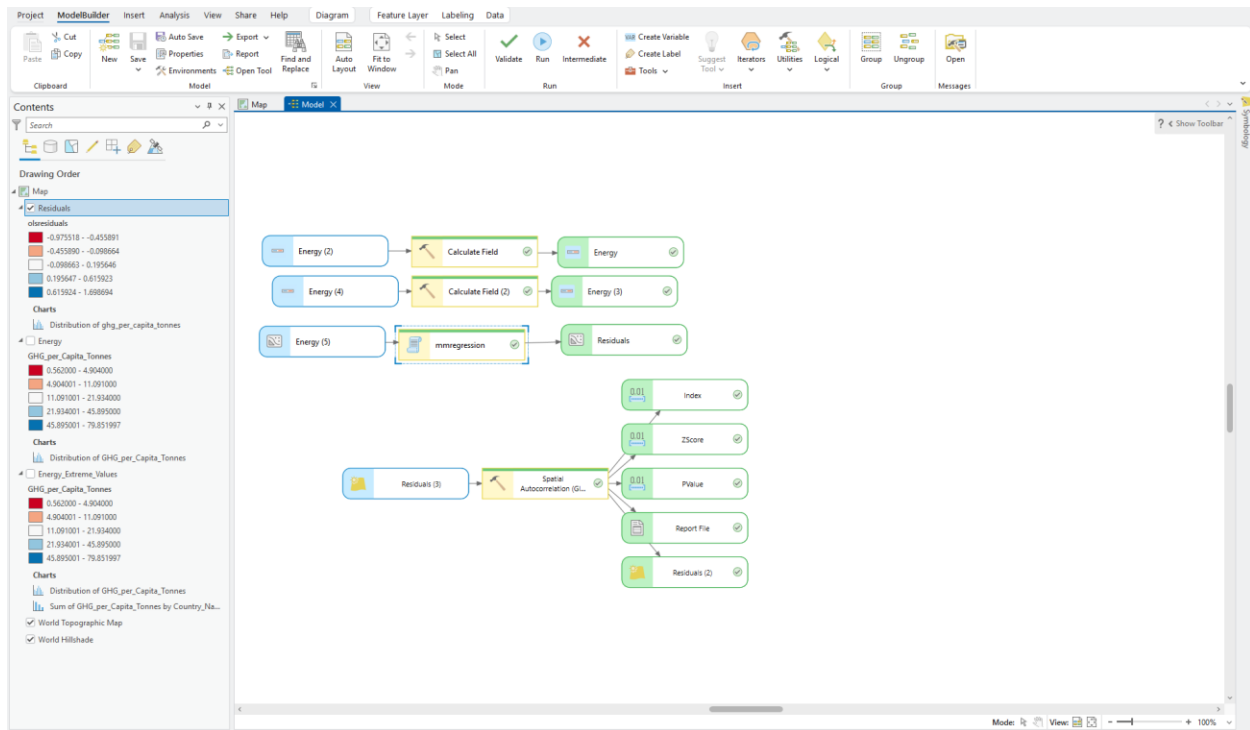
The Global Moran's z-score of 3.30 indicates that the residuals are spatially autocorrelated.



Gas consumption per capita has an OLS coefficient value of 0.5124 and a spatial lag coefficient value of 0.476.



Gas consumption per capita has an OLS coefficient value of 0.5124 and spatial error coefficient value of 0.478



ModelBuilder